Model Development Phase Template

Date	15 March 2024
Team ID	LTVIP2026TMIDS76529
Project Title	Advancing Nutrition Science through GeminiAI (NutriGen)
Maximum Marks	4 Marks

Initial Model Training Code, Model Validation and Evaluation Report

NutriGen is developed using Google Generative AI (Gemini API) to generate personalized meal plans, nutritional insights, and virtual coaching responses. Instead of traditional machine learning classification models, the system integrates a pre-trained large language model to generate intelligent outputs based on user prompts.

Paste the screenshot of the model training code

Model Validation and Evaluation Report:

Model Evaluation Table:

Model	Model Description	Classification Report	Accuracy / Evaluation Metric	Confusion Matrix
Model 1	Personalized Meal Plan Generation	Not Applicable (Generative AI based system)	Evaluated based on nutritional balance and user alignment	Not Applicable
Model 2	Dynamic Nutritional Insight Generation	Not Applicable	Evaluated based on correctness of macro, micro & calories	Not Applicable
Model 3	Virtual Nutrition Coaching	Not Applicable	Evaluated based on relevance, personalization & consistency	Not Applicable

Validation Testing Summary Table:

Test Case	Input Condition	Expected Output	Result
Case 1	Vegetarian + Weight Loss	Low calorie vegetarian meal plan	Successfully Generated
Case 2	Diabetic User	Low sugar balanced meal plan	Successfully Generated
Case 3	High Activity Level	High protein meal recommendation	Successfully Generated
Case 4	Peanut Allergy	Exclusion of peanutbased items	Successfully Handled
Case 5	General Nutrition Query	Accurate nutrient breakdown	Successfully Generated

Evaluation Method Table:

Evaluation Type	Method Used
Functional Testing	API response validation and structured output checking
Personalization Testing	Tested with multiple user profiles
Nutritional Accuracy Testing	Verified logical nutrient distribution
Response Quality Testing	Checked contextual and relevant AI responses